

Representative Structural Calculation Package

Courtyard hospital frame

Item	Value
Project	Courtyard hospital frame
Purpose	Representative elastic analysis and preliminary design trace
Generated	2026-06-27T00:00:00.000Z
Package version	m42-calculation-package
Source model	08-courtyard-hospital

Nodes	Members	Loads	Combos	Max disp.	Max util.
120	240	672	28	9.646 mm	0.628

Model And Design Basis

Item	Value
Building type	hospital
Description	Four-story public facility frame with a central courtyard opening.
Occupancy	Hospital
Bounds X/Y/Z	24.00 / 24.00 / 15.20
Analysis status	OK
Design status	OK
Package audit	OK

Load Case Trace

Case	Type	Loads	Fx	Fy	Fz
D	dead	144	0.000 kN	0.000 kN	-8525 kN
L	live	144	0.000 kN	0.000 kN	-4032 kN
WX	wind	96	164.16 kN	0.000 kN	0.000 kN
WY	wind	96	0.000 kN	164.16 kN	0.000 kN
EX	seismic	96	753.09 kN	0.000 kN	0.000 kN
EY	seismic	96	0.000 kN	753.09 kN	0.000 kN

Load Derivation

Story	Area	D total	L total	Wind X	Wind Y
1	576.00	2131 kN	1152 kN	41.04 kN	41.04 kN
2	576.00	2131 kN	1152 kN	41.04 kN	41.04 kN
3	576.00	2131 kN	1152 kN	41.04 kN	41.04 kN
4	576.00	2131 kN	576.00 kN	41.04 kN	41.04 kN

Load Derivation Formula Trace

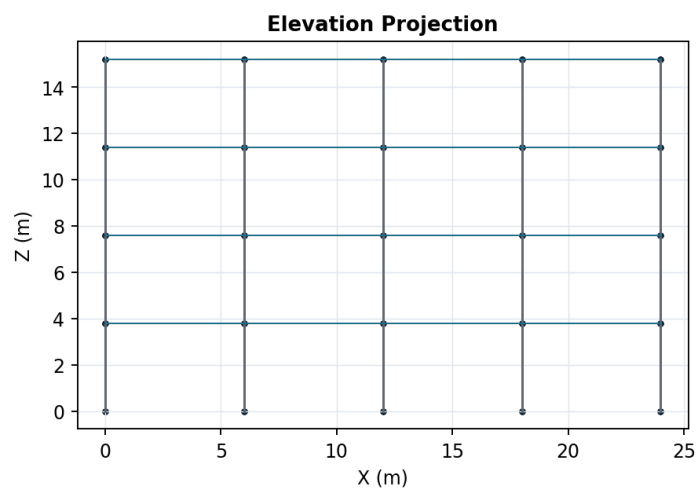
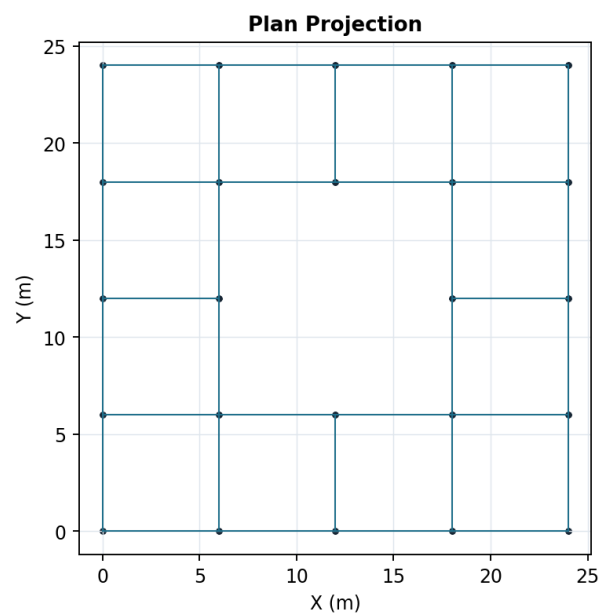
ID	Case	Formula	Result	Inputs
basis-occupancy	-	occupancy preset lookup	-	occupancy=hospital
basis-plan-area	-	$\max(\text{model width} * \text{model depth}, 1)$	576.00 m2	Bx=24.00 m, By=24.00 m
basis-roof-area	-	roofArea input or typical floor area	576.00 m2	roofArea=576.00 m2
D-ST1	D	$A * q_D$	2131 kN	$A=576.00 \text{ m}^2$, $q_D=3.700 \text{ kN/m}^2$
L-ST1	L	$A * q_L$	1152 kN	$A=576.00 \text{ m}^2$, $q_L=2.000 \text{ kN/m}^2$
WX-ST1	WX	$p_{WX} * B_y * h$	41.04 kN	$p_{WX}=0.450 \text{ kN/m}^2$, $B_y=24.00 \text{ m}$, $h=3.800 \text{ m}$
WY-ST1	WY	$p_{WY} * B_x * h$	41.04 kN	$p_{WY}=0.450 \text{ kN/m}^2$, $B_x=24.00 \text{ m}$, $h=3.800 \text{ m}$
D-ST2	D	$A * q_D$	2131 kN	$A=576.00 \text{ m}^2$, $q_D=3.700 \text{ kN/m}^2$
L-ST2	L	$A * q_L$	1152 kN	$A=576.00 \text{ m}^2$, $q_L=2.000 \text{ kN/m}^2$
WX-ST2	WX	$p_{WX} * B_y * h$	41.04 kN	$p_{WX}=0.450 \text{ kN/m}^2$, $B_y=24.00 \text{ m}$, $h=3.800 \text{ m}$
WY-ST2	WY	$p_{WY} * B_x * h$	41.04 kN	$p_{WY}=0.450 \text{ kN/m}^2$, $B_x=24.00 \text{ m}$, $h=3.800 \text{ m}$
D-ST3	D	$A * q_D$	2131 kN	$A=576.00 \text{ m}^2$, $q_D=3.700 \text{ kN/m}^2$
L-ST3	L	$A * q_L$	1152 kN	$A=576.00 \text{ m}^2$, $q_L=2.000 \text{ kN/m}^2$
WX-ST3	WX	$p_{WX} * B_y * h$	41.04 kN	$p_{WX}=0.450 \text{ kN/m}^2$, $B_y=24.00 \text{ m}$, $h=3.800 \text{ m}$

KDS-Style Load Standard Audit

Symbol	Status	Mapped cases	Project input
D	mapped	D	Optional
L	mapped	L	Optional
Lr	missing	-	Optional
S	missing	-	Required
R	missing	-	Required
W	mapped	WX, WY	Required
E	mapped	EX, EY	Required
H	missing	-	Required
F	missing	-	Required
T	missing	-	Required

Preset	Status	Missing	Missing any	Generated
KDS-ST-03	blocked	-	Lr, S, R	0

Courtyard hospital frame



Combination Analysis Results

Combo	Status	Max disp.	Max util.	Residual
KDS-ST-01	OK	1.648 mm	0.177	0.000
KDS-ST-02	OK	2.118 mm	0.248	0.000
KDS-ST-04-WX-P	OK	2.497 mm	0.247	0.000
KDS-ST-04-WX-N	OK	2.497 mm	0.247	0.000
KDS-ST-04-WY-P	OK	2.497 mm	0.247	0.000
KDS-ST-04-WY-N	OK	2.497 mm	0.247	0.000
KDS-ST-05-EX-P	OK	9.646 mm	0.385	0.000
KDS-ST-05-EX-N	OK	9.646 mm	0.385	0.000
KDS-ST-05-EY-P	OK	9.646 mm	0.385	0.000
KDS-ST-05-EY-N	OK	9.646 mm	0.385	0.000
KDS-ST-06-WX-P	OK	2.001 mm	0.150	0.000
KDS-ST-06-WX-N	OK	2.001 mm	0.150	0.000
KDS-ST-06-WY-P	OK	2.001 mm	0.150	0.000
KDS-ST-06-WY-N	OK	2.001 mm	0.150	0.000
KDS-ST-07-EX-P	OK	9.513 mm	0.288	0.000
KDS-ST-07-EX-N	OK	9.513 mm	0.288	0.000

Serviceability Drift Review

Item	Value
Limit	H/200
Status	OK
Max drift	3.144 mm
Max drift ratio	0.00083
Governing	KDS-ST-05-EX-P / story 2

Combo	Story	Height	Drift	Ratio	D/L	Status
KDS-ST-01	1	3.800 m	0.015 mm	0.00000	0.001	OK
KDS-ST-01	2	3.800 m	0.016 mm	0.00000	0.001	OK
KDS-ST-01	3	3.800 m	0.015 mm	0.00000	0.001	OK
KDS-ST-01	4	3.800 m	0.060 mm	0.00002	0.003	OK
KDS-ST-02	1	3.800 m	0.021 mm	0.00001	0.001	OK
KDS-ST-02	2	3.800 m	0.022 mm	0.00001	0.001	OK
KDS-ST-02	3	3.800 m	0.015 mm	0.00000	0.001	OK
KDS-ST-02	4	3.800 m	0.070 mm	0.00002	0.004	OK
KDS-ST-04-WX-P	1	3.800 m	0.466 mm	0.00012	0.025	OK
KDS-ST-04-WX-P	2	3.800 m	0.588 mm	0.00015	0.031	OK
KDS-ST-04-WX-P	3	3.800 m	0.431 mm	0.00011	0.023	OK
KDS-ST-04-WX-P	4	3.800 m	0.283 mm	0.00007	0.015	OK
KDS-ST-04-WX-N	1	3.800 m	0.466 mm	0.00012	0.025	OK
KDS-ST-04-WX-N	2	3.800 m	0.588 mm	0.00015	0.031	OK
KDS-ST-04-WX-N	3	3.800 m	0.431 mm	0.00011	0.023	OK
KDS-ST-04-WX-N	4	3.800 m	0.283 mm	0.00007	0.015	OK
KDS-ST-04-WY-P	1	3.800 m	0.466 mm	0.00012	0.025	OK
KDS-ST-04-WY-P	2	3.800 m	0.588 mm	0.00015	0.031	OK
KDS-ST-04-WY-P	3	3.800 m	0.431 mm	0.00011	0.023	OK
KDS-ST-04-WY-P	4	3.800 m	0.283 mm	0.00007	0.015	OK

Governing Members

Rank	Member	Status	Util.	Check	Combo
1	M85	OK	0.628	steel-deflection	KDS-ST-05-EY-P
2	M84	OK	0.628	steel-deflection	KDS-ST-05-EY-N
3	M89	OK	0.628	steel-deflection	KDS-ST-05-EX-P
4	M80	OK	0.628	steel-deflection	KDS-ST-05-EX-P
5	M86	OK	0.627	steel-deflection	KDS-ST-05-EY-N
6	M83	OK	0.627	steel-deflection	KDS-ST-05-EY-P
7	M94	OK	0.627	steel-deflection	KDS-ST-05-EX-N
8	M75	OK	0.627	steel-deflection	KDS-ST-05-EX-N
9	M81	OK	0.612	steel-deflection	KDS-ST-05-EX-N
10	M79	OK	0.612	steel-deflection	KDS-ST-05-EX-P

11	M90	OK	0.612	steel-deflection	KDS-ST-05-EX-N
12	M88	OK	0.612	steel-deflection	KDS-ST-05-EY-N
13	M82	OK	0.609	steel-deflection	KDS-ST-05-EY-P
14	M78	OK	0.609	steel-deflection	KDS-ST-05-EY-P
15	M91	OK	0.609	steel-deflection	KDS-ST-05-EY-N
16	M87	OK	0.609	steel-deflection	KDS-ST-05-EY-N
17	M74	OK	0.609	steel-deflection	KDS-ST-05-EX-P
18	M93	OK	0.609	steel-deflection	KDS-ST-05-EX-P
19	M76	OK	0.609	steel-deflection	KDS-ST-05-EX-N
20	M95	OK	0.609	steel-deflection	KDS-ST-05-EX-N

Member Design Trace Matrix

Item		Value			
Checked members		240			
Unimplemented members		0			
NG / WARN		0/0			
Max utilization		0.628			

Member	Type	Status	Util.	Formula	Action
M85	steel	OK	0.628	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M84	steel	OK	0.628	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M89	steel	OK	0.628	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M80	steel	OK	0.628	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M86	steel	OK	0.627	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M83	steel	OK	0.627	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M94	steel	OK	0.627	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M75	steel	OK	0.627	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M81	steel	OK	0.612	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M79	steel	OK	0.612	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.

M90	steel	OK	0.612	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M88	steel	OK	0.612	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M82	steel	OK	0.609	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M78	steel	OK	0.609	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M91	steel	OK	0.609	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M87	steel	OK	0.609	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M74	steel	OK	0.609	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M93	steel	OK	0.609	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M76	steel	OK	0.609	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M95	steel	OK	0.609	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.

Steel, Connection, Foundation Summary

Scope	Rows	Warnings/NG	Max ratio
Steel review	240	0	0.628
Connections	240	72	3.716
Foundations	24	72	1.000

Quality Audit

Audit item	Status
Load derivation attached	OK
Combination results available	OK
Member checks available	OK
RC schedule available	Not applicable
Steel schedule available	OK
Foundation review available	OK

Remaining Scope

- Project design basis: occupancy, importance factor, site class, exposure, wind/seismic procedure, and serviceability criteria.
- Load derivation: dead, live, wind, seismic, snow, rain, soil, temperature, and construction load calculation sheets.

- Full KDS equation trace for every generated load combination and project-specific exception.
- RC member detailing: longitudinal bars, stirrups, development length, lap splice, anchorage, and constructability checks.
- Steel member detailing: compactness, lateral torsional buckling, connection forces, base plates, and weld/bolt checks.
- Foundation design: soil bearing, pile/footing design, settlement, uplift, sliding, overturning, and reinforcement.
- Drawing import and vision-assisted modeling audit trail for future CAD/image/MGT workflows.
- Drawing/image/MGT import audit trail is not attached to this generated test set.